Section 4.9 Integration

If $y = x^3$, then $y' = 3x^2$.

If $y' = \frac{x^{7+1}}{8}$, then $y = \frac{x^8}{8} + c$ ← constant of integration

S $\int x^7 \cdot dx$

↖ integral symbol

Ex ① $\int x^2\, dx = \frac{x^{2+1}}{3} + c = \boxed{\frac{x^3}{3} + c}$

② $\int e^x\, dx = \boxed{e^x + c}$

③ $\int \frac{1}{x}\, dx = \boxed{\ln x + c}$

(4) $\int \cos x \, dx = \boxed{\sin x + c}$

(5) $\int \sin x \, dx = \boxed{-\cos x + c}$

(6) $\int \frac{1}{\cos^2 x} \, dx = \int \sec^2 x \, dx = \boxed{\tan x + c}$

(7) $\int \sec x \cdot \tan x \, dx = \boxed{\sec x + c}$

(8) $\int \frac{1}{1+x^2} \, dx = \boxed{\tan^{-1} x + c}$

(9) $\int 100 \, dx = \boxed{100x + c}$

(10) $\int \frac{x+7}{x} \, dx = \int \left(\frac{x}{x} + \frac{7}{x}\right) \cdot dx$

$= \int \left(1 + \frac{7}{x}\right) dx = \boxed{x + 7 \cdot \ln x + c}$

Ex (11) Solve the differential equation

$$\frac{dy}{dt} = \frac{1}{t^2} + \frac{t}{4} \quad \text{given} \quad y(2) = 1$$

$$dt\,\frac{dy}{dt} = \left(t^{-2} + \frac{t}{4}\right) dt$$

$$\int dy = \int \left(\frac{t^{-2+1}}{-1} + \frac{t^{1+1}}{4\cdot 2}\right) dt$$

$$y = -\frac{1}{t} + \frac{t^2}{8} + c$$

$$1 = -\frac{1}{2} + \frac{2^2}{8} + c$$

$$1 = c$$

$$\boxed{y = -\frac{1}{t} + \frac{t^2}{8} + 1}$$

Ex (12) $\int \sqrt[3]{x+1}\, dx$

$$= \int \frac{(x+1)^{1/3+1}}{\frac{4}{3}}\, dx$$

$$= \boxed{\frac{3}{4}(x+1)^{4/3} + c}$$

Ex (13) $\int e^{7x} \cdot dx$

$$= \boxed{\frac{e^{7x}}{7} + c}$$

Ex (14) $\int \cos(14x)\, dx$

$$= \boxed{\frac{\sin(14x)}{14} + c}$$